

```
In [1]: import pandas as pd
        from sklearn import preprocessing
```

```
In [2]: df = pd.read_csv('Datasets/Mall_Customers.csv')
```

```
In [3]: df
```

Out[3]:

	CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40
...	...	...	...	...	...
195	196	Female	35	120	79
196	197	Female	45	126	28
197	198	Male	32	126	74
198	199	Male	32	137	18
199	200	Male	30	137	83

200 rows × 5 columns

```
In [4]: df.describe()
```

Out[4]:

	CustomerID	Age	Annual Income (k\$)	Spending Score (1-100)
count	200.000000	200.000000	200.000000	200.000000
mean	100.500000	38.850000	60.560000	50.200000
std	57.879185	13.969007	26.264721	25.823522
min	1.000000	18.000000	15.000000	1.000000
25%	50.750000	28.750000	41.500000	34.750000
50%	100.500000	36.000000	61.500000	50.000000
75%	150.250000	49.000000	78.000000	73.000000
max	200.000000	70.000000	137.000000	99.000000

```
In [5]: df.min()
```

Out[5]: CustomerID 1
Gender Female
Age 18
Annual Income (k\$) 15
Spending Score (1-100) 1
dtype: object

```
In [6]: df.groupby(['Gender'])['Age'].mean()
```

Out[6]: Gender
Female 38.098214
Male 39.806818
Name: Age, dtype: float64

```
In [7]: df.groupby(['Gender'])['Age'].median()
```

Out[7]: Gender
Female 35.0
Male 37.0
Name: Age, dtype: float64

```
In [8]: df.groupby(['Gender'])['Age'].std()
```

Out[8]: Gender
Female 12.644095
Male 15.514812
Name: Age, dtype: float64

```
In [9]: df.groupby(['Gender'])['Annual Income (k$)'].mean()
```

Out[9]: Gender
Female 59.250000
Male 62.227273
Name: Annual Income (k\$), dtype: float64

```
In [10]: df.groupby(['Gender'])['Annual Income (k$)'].median()
```

Out[10]: Gender
Female 60.0
Male 62.5
Name: Annual Income (k\$), dtype: float64

```
In [11]: df.groupby(['Gender'])['Annual Income (k$)'].std()
```

Out[11]: Gender
Female 26.011952
Male 26.638373
Name: Annual Income (k\$), dtype: float64

```
In [12]: df.groupby(['Gender']).mean()
```

Out[12]:

	CustomerID	Age	Annual Income (k\$)	Spending Score (1-100)
Gender				
Female	97.562500	38.098214	59.250000	51.526786
Male	104.238636	39.806818	62.227273	48.511364

```
In [13]: df.groupby(['Gender']).median()
```

Out[13]:

	CustomerID	Age	Annual Income (k\$)	Spending Score (1-100)
Gender				
Female	94.5	35.0	60.0	50.0
Male	106.5	37.0	62.5	50.0

```
In [14]: df.groupby(['Gender']).min()
```

Out[14]:

	CustomerID	Age	Annual Income (k\$)	Spending Score (1-100)
Gender				
Female	3	18	16	5
Male	1	18	15	1

```
In [15]: df.groupby(['Gender']).max()
```

Out[15]:

	CustomerID	Age	Annual Income (k\$)	Spending Score (1-100)
Gender				
Female	197	68	126	99
Male	200	70	137	97

```
In [16]: x = df.drop(axis=1, columns=['Gender'])
x
```

Out[16]:

	CustomerID	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	19	15	39
1	2	21	15	81
2	3	20	16	6
3	4	23	16	77
4	5	31	17	40
...	...	...	...	...
195	196	35	120	79
196	197	45	126	28
197	198	32	126	74
198	199	32	137	18
199	200	30	137	83

200 rows × 4 columns

```
In [17]: enc = preprocessing.OneHotEncoder()
enc_df = pd.DataFrame(enc.fit_transform(df[['Gender']]).toarray())
enc_df
```

Out[17]:

	0	1
0	0.0	1.0
1	0.0	1.0
2	1.0	0.0
3	1.0	0.0
4	1.0	0.0
...	...	...
195	1.0	0.0
196	1.0	0.0
197	0.0	1.0
198	0.0	1.0
199	0.0	1.0

200 rows × 2 columns

```
In [18]: df_encode = x.join(enc_df)
df_encode
```

Out[18]:

	CustomerID	Age	Annual Income (k\$)	Spending Score (1-100)	0	1
0	1	19	15	39	0.0	1.0
1	2	21	15	81	0.0	1.0
2	3	20	16	6	1.0	0.0
3	4	23	16	77	1.0	0.0
4	5	31	17	40	1.0	0.0
...	...	...	...	...	...	...
195	196	35	120	79	1.0	0.0
196	197	45	126	28	1.0	0.0
197	198	32	126	74	0.0	1.0
198	199	32	137	18	0.0	1.0
199	200	30	137	83	0.0	1.0

200 rows × 6 columns

```
In [19]: df1 = pd.read_csv('Datasets/Iris.csv')
```

```
In [20]: df1
```

Out[20]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...	...
145	146	6.7	3.0	5.2	2.3	Iris-virginica
146	147	6.3	2.5	5.0	1.9	Iris-virginica
147	148	6.5	3.0	5.2	2.0	Iris-virginica
148	149	6.2	3.4	5.4	2.3	Iris-virginica
149	150	5.9	3.0	5.1	1.8	Iris-virginica

150 rows × 6 columns

In [21]:

df1.describe()

Out[21]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.843333	3.054000	3.758667	1.198667
std	43.445368	0.828066	0.433594	1.764420	0.763161
min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000
50%	75.500000	5.800000	3.000000	4.350000	1.300000
75%	112.750000	6.400000	3.300000	5.100000	1.800000
max	150.000000	7.900000	4.400000	6.900000	2.500000

In [22]:

set1 = (df1['Species'] == 'Iris-virginica')

In [23]:

print(df1[set1].describe())

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	50.00000	50.00000	50.00000	50.00000	50.00000
mean	125.50000	6.58800	2.97400	5.55200	2.02600
std	14.57738	0.63588	0.32249	0.55189	0.27465
min	101.00000	4.90000	2.20000	4.50000	1.40000
25%	113.25000	6.22500	2.80000	5.10000	1.80000
50%	125.50000	6.50000	3.00000	5.55000	2.00000
75%	137.75000	6.90000	3.17500	5.87500	2.30000
max	150.00000	7.90000	3.80000	6.90000	2.50000

In [24]:

set2 = (df1['Species'] == 'Iris-versicolor')

In [25]:

print(df1[set2].describe())

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	50.00000	50.000000	50.000000	50.000000	50.000000
mean	75.50000	5.936000	2.770000	4.260000	1.326000
std	14.57738	0.516171	0.313798	0.469911	0.197753
min	51.00000	4.900000	2.000000	3.000000	1.000000
25%	63.25000	5.600000	2.525000	4.000000	1.200000
50%	75.50000	5.900000	2.800000	4.350000	1.300000
75%	87.75000	6.300000	3.000000	4.600000	1.500000
max	100.00000	7.000000	3.400000	5.100000	1.800000

```
In [26]: set3 = (df1['Species'] == 'Iris-setosa')
```

```
In [27]: print(df1[set3].describe())
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	50.00000	50.00000	50.000000	50.000000	50.00000
mean	25.50000	5.00600	3.418000	1.464000	0.24400
std	14.57738	0.35249	0.381024	0.173511	0.10721
min	1.00000	4.30000	2.300000	1.000000	0.10000
25%	13.25000	4.80000	3.125000	1.400000	0.20000
50%	25.50000	5.00000	3.400000	1.500000	0.20000
75%	37.75000	5.20000	3.675000	1.575000	0.30000
max	50.00000	5.80000	4.400000	1.900000	0.60000

```
In [28]: df1['Species'].unique()
```

```
Out[28]: array(['Iris-setosa', 'Iris-versicolor', 'Iris-virginica'], dtype=object)
```

```
In [29]: df1.groupby(['Species']).mean()
```

```
Out[29]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
Species					
Iris-setosa	25.5	5.006	3.418	1.464	0.244
Iris-versicolor	75.5	5.936	2.770	4.260	1.326
Iris-virginica	125.5	6.588	2.974	5.552	2.026

```
In [30]: df1.groupby(['Species']).median()
```

```
Out[30]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
Species					
Iris-setosa	25.5	5.0	3.4	1.50	0.2
Iris-versicolor	75.5	5.9	2.8	4.35	1.3
Iris-virginica	125.5	6.5	3.0	5.55	2.0

```
In [31]: df1.groupby(['Species']).std()
```

Out[31]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
Species					
Iris-setosa	14.57738	0.352490	0.381024	0.173511	0.107210
Iris-versicolor	14.57738	0.516171	0.313798	0.469911	0.197753
Iris-virginica	14.57738	0.635880	0.322497	0.551895	0.274650